

Report
Of
NLP-Based Customer Support System

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: CS-2341020

Name of Student: ANSHUMAN SINGH

Batch: 2023-2027

CANDIDATE'S DECLARATION

I, Anshuman Singh do hereby declare that the internship report titled NLP-Based Customer Support has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature Anshuman singh

Student Name Anshuman

Singh

Roll No. CS-2341020

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Pratinik Infotech for providing me with the opportunity to work as an ML Engineer Intern. I am thankful to my mentors and colleagues for their guidance, support, and valuable feedback throughout the internship. I also thank my university faculty and my family for their constant encouragement and support.

Signature Anshuman singh

Student Name Anshuman

Singh

Roll No. CS-2341020

4. PROJECT DESCRIPTION

4.1 Introduction

During my internship at Pratinik Infotech, I worked as an ML Engineer Intern. The main project developed during the internship was an NLP-Based Customer Support System. The purpose of the project was to use Natural Language Processing to understand customer queries and provide relevant, efficient, and automated responses.

4.2 Organization Profile

Pratinik Infotech is a technology-focused organization that provides software and digital solutions. During my internship, I gained practical exposure to machine learning concepts, Natural Language Processing, data processing, and the development of AI-based applications

4.3 Problem Statement

Customer support systems often receive a large number of repetitive queries. Providing manual responses to every query can be time-consuming and inefficient. The project aimed to develop an NLP-based system capable of understanding customer queries and generating or retrieving relevant responses.

4.4 Project Objectives

The main objectives of the project were:

- To develop an NLP-based customer support system.
- To preprocess and analyze customer queries.
- To classify or understand user intent.
- To provide relevant responses automatically.
- To improve the speed and efficiency of customer support.

4.5 Scope of the Project

The system can be used to automate responses to common customer queries and provide basic support through an intelligent conversational interface. It can be further enhanced by integrating larger datasets, advanced language models, multilingual support, and real-time deployment.

4.6 Technologies and Tools Used

The project involved the use of the following technologies and tools:

- Python
- Natural Language Processing (NLP)
- Machine Learning
- Pandas and NumPy
- Scikit-learn
- Jupyter Notebook / Google Colab
- NLP libraries such as NLTK or spaCy

4.7 System Architecture

The system architecture consists of the following stages:

Customer Query → Text Preprocessing → Feature Extraction / NLP
Processing → ML Model or Intent Detection → Response
Generation → Customer Response

The system receives a customer query, processes the text, identifies the relevant intent or information, and provides an appropriate response.

4.8 Methodology

The project was developed using the following methodology:

1. Collection or preparation of customer support data.
2. Cleaning and preprocessing of textual data.
3. Tokenization and normalization of text.
4. Feature extraction and NLP processing.
5. Training and testing of the machine learning model.
6. Evaluating model performance.
7. Integrating the model into the customer support workflow.

4.9 Expected Outcomes

The expected outcomes of the project include:

- Automated handling of common customer queries.
- Faster customer response times.
- Reduced manual effort in customer support.
- Improved consistency in responses.
- Practical implementation of NLP and machine learning concepts.

The internship helped me gain practical experience in applying Machine Learning and Natural Language Processing techniques to solve real-world problems.

4.10 Internship Completion Certificate



4.10 Internship Offer Letter



Internship Offer Letter

Date: 10/06/2026
Intern Id: 6172

Dear Anshuman Singh,

We are pleased to offer you the opportunity to join **Pratinik Infotech** as an Intern. We were impressed with your qualifications and your passion for **Machine Learning**, and we believe that you will make a valuable contribution to our team.

Position: Machine Learning Intern

Duration: 10th June 2026 to 9th August 2026

We look forward to welcoming you to the **Pratinik Infotech** team and working together to achieve success. Thank you for choosing **Pratinik Infotech** as the place to further your career and professional growth.


Sincerely,
HR Manager
Pratinik Infotech
www.pratinik.com

info@pratinik.com | Pratinik Infotech | www.pratinik.com

5. BIBLIOGRAPHY / REFERENCES

1. Python Documentation.
2. Scikit-learn Documentation.
3. NLTK Documentation.
4. Relevant books and research papers on Natural Language Processing and Machine Learning.
5. Official documentation of the libraries and tools used during the project.